

P1 – Mitigating Public Transit Unreliability in Malaysia through Deep Learning Analytics and Ridership Forecasting

Master of Artificial Intelligence



UNIVERSITI
MALAYA

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Research Background

Public transportation is **essential** for **urban mobility**, **economic growth**, and **environmental sustainability**.

Rapid urbanization in Malaysia **increases demand** for **efficient** public transit systems.

Reliable public transit, such as **buses** and **rails**, improves **accessibility** and **quality of life** by being **punctual** and **efficient**.





01

Literature Review

Public Transport Reliability & Demand Dynamics

J. Articles	Focus	Method	Dataset	Strength	Weakness
Ibrahim et al. (2022)	Investigate service quality factors influencing user satisfaction with LRT in Klang Valley.	Factor analysis, correlation, and Artificial Neural Network (ANN).	417 LRT riders in Klang Valley, Malaysia.	Uses ANN for non-linear modeling; local Malaysian context.	Focuses on satisfaction/perceptions, not operational ridership management.
Mee & Subramaniam (2022)	Examine willingness to use public transport in KL and Manila.	Purposive sampling survey; PLS-SEM analysis.	209 respondents from KL and Manila.	Comparative city analysis; identifies key behavioral drivers.	Small sample size; focus on "willingness" rather than actual ridership data.
Rezapour & Ferraro (2021)	Impact of rail delays and real-time info accuracy on passenger psychology.	Survey-based; Factor Analysis and SEM.	396 KTM rail passengers at Serdang station, Malaysia.	Links psychological impact to information systems; local KTM context.	Subjective measures; does not propose operational management strategies.
Ubaidillah & Syafiq (2022)	Impact of bus service quality on satisfaction in a developing city.	SERVQUAL model; PLS-SEM analysis.	300 bus users in Kota Samarahan, Sarawak, Malaysia.	Focuses on a developing city context with high car ownership.	Relies on self-reported data; no integration of actual ridership metrics.

Transit Data Ecosystems & Anomalies

J. Articles	Focus	Method	Dataset	Strength	Weakness
Keller et al. (2022)	Potential of data science for sustainable public transport.	Systematic review and expert interviews.	Global/Germany ; Multi-source big data (AFC, APC, GPS).	Comprehensive taxonomy of data sources and use cases.	Generic framework; lacks specific application to the Malaysian context.
Liu et al. (2025)	Survey of vehicle trajectory data processing and analytics.	Systematic review of collection and processing methods.	Global (China, US, SE Asia); Taxi, bus, and truck trajectories.	Deep dive into trajectory collection and privacy-preserving methods.	Survey-based; no direct implementation for Malaysian transit.
Lu et al. (2020)	Review of big data applications in urban transit systems.	Literature review of 95 global cases.	Global multi-source transit data (AFC, AVL, APC, GTFS).	Synthesizes heterogeneous data sources and their management roles.	Does not provide an empirical model or Malaysian case study.
Tu et al. (2021)	Review of transit data sources and their complementarity.	Literature review; Information management framework.	Global; AVL, AFC, APC, and exogenous data (weather).	Highlights the value of data fusion (e.g., weather + transit).	Conceptual; no direct implementation or Malaysian validation.

Evolution from Traditional to DL Models

J. Articles	Focus	Method	Dataset	Strength	Weakness
Jiang et al. (2021)	Survey and benchmark of deep learning for traffic prediction.	Survey and comparative experiments.	Global public traffic datasets (METR-LA, PEMS-BAY).	Comprehensive taxonomy and experimental benchmarking.	No direct application to public transit ridership or Malaysia.
Sangeetha & Kiros (2024)	GeoTemporal LSTM for predictive modeling of urban mobility.	GT-LSTM combining Attention and LSTM.	NYC Taxi, GTFS, and OSM data.	Innovative fusion of spatial and temporal features with attention.	High complexity; tested only on NYC data; not Malaysian-specific.
Sobrie et al. (2023)	Real-time train delay prediction using deep learning.	LSTM encoder-decoder with spatiotemporal features.	Belgian railway network; 3M+ signal passings/month.	Large-scale real-world operationalization; captures network complexity.	Belgium-specific; focus on delays rather than ridership volume.
Yin et al. (2021)	Deep learning on traffic prediction: Methods and analysis.	Survey and taxonomy of deep learning models.	Global traffic datasets; Road network topology.	Thorough analysis of spatio-temporal dependency modeling.	Survey-based; lacks specific transit ridership management focus.

AI-Driven Operational Strategies & Sustainability

J. Articles	Focus	Method	Dataset	Strength	Weakness
Gheorghe & Șoica (2025)	Review of AI, IoT, and predictive analytics in traffic control.	Systematic literature review (2020-2024).	Global; Smart city traffic management systems.	Up-to-date synthesis of adaptive traffic control technologies.	No empirical validation or data from Malaysia's transit network.
Jevinger & Persson (2024)	AI for improving public transport: A mapping study.	Systematic mapping study of 111 primary studies.	Global; Bus, train, and multimodal systems.	Up-to-date synthesis of adaptive traffic control technologies.	Lacks localized solutions or models for the Malaysian context.
Sobrie et al. (2023)	Real-time train delay prediction comparing rule-based vs. deep learning models.	Customized LSTM encoder-decoder deep learning model.	Real-world network-wide Belgian railway data (Infrabel).	Demonstrates that DL superiority increases with system complexity; implemented as a real-world Proof of Concept.	High dependency on real-time data quality and lacks the transparency of rule-based models.
Son & Kim (2025)	Leveraging advanced technologies for transportation planning.	Systematic review using Sentence-BERT NLP.	Global (Urban, Rural, and Simulated contexts).	Innovative use of NLP for semantic analysis of research trends.	No primary data or application to Malaysia's transit network.



02

Gapping the Problem

Research Gaps are identified as below:

Limited Integration of Multi-scale Spatio-temporal Features in Malaysia

Inadequate Quantification and Mitigation of Extreme Anomaly Bias

Lack of Comprehensive Benchmarking Across SOTA Architectures





Research Problem Statement

The difficulty in accurately predicting daily ridership due to complex spatiotemporal factors (population density, weather, accessibility, etc.) **(Amir et al., 2025)**.

This inability to predict ridership leads to critical bottlenecks in scheduling, routing, and service intervals during peak hours and holidays, harming system reliability and passenger satisfaction.

Historical ridership data is contaminated by the extreme 85–90% passenger drop caused by the MCO period **(Amir et al., 2025)**. Training AI models on this data introduces severe predictive bias **(Zhao & Lan, 2025)** and prevents them from recognizing true ridership peaks after normal life resumes **(Lv et al., 2021)**.



03

Research Questions & Objectives

Research Questions & Objectives

Research Questions	Research Objectives
What are the primary spatial and temporal factors (e.g. weather, POIs, population density) that heavily affect public transit ridership trends in Malaysia?	To identify and categorize the primary spatial and temporal factors affecting Malaysian public transit ridership through a systematic analysis of historical performance data.
How accurately and consistently can existing DL architectures (spatio-temporal, graph-based, and attention-based) compare against each other and proposed model in predicting Malaysia's public transit ridership?	To develop Hybrid Multi-scale Temporal Spatio-Feature (HMT-TSF) Forecaster that integrates Malaysia's spatial and temporal data to estimate future ridership trends.
How does the inclusion of extreme data anomalies, specifically the Malaysia's MCO period, impact the predictive accuracy and bias of existing DL models and proposed model?	To benchmark the proposed model against existing baseline, spatio-temporal-based, graph-based and attention-based models using combined MAE, RMSE, MAPE, and R^2 .
	To evaluate and compare model performance across two distinct dataset segmentations (with and without MCO period) to determine the impact of extreme outlier data on predictive bias.



04

Research Scope

Research Scope



Models Compared

Baseline LSTM, 4 spatio-temporal models, 5 graph models, and 4 attention models.



Evaluation Metrics

Combined MAE, RMSE, and MAPE as overall accuracy metrics & R^2



Geographical Focus

Bus, urban rail, commuter transit networks with available data throughout whole Malaysia.



MCO Period Segmentation

Without MCO segment: 1 Jan 2022 – 31 Dec 2025

With MCO segment: 1 Jan 2019 – 31 Dec 2025



05a

How I Execute

Data Sources

Data	Data Origin	Why Data?	References
Ridership	data.gov.my	Primary forecast target, provides 13 output variables (12 service lines + total) the models are trained to predict	Khalil et al., 2021; Amir et al., 2025
Fuel Price	data.gov.my	Demand driver, EDA confirms higher fuel prices correlate with increased transit use, capturing modal substitution behaviour	Ahmad et al., 2024; Cui et al., 2025
Holidays	timeanddate.com	Demand modifier, holidays cause systematic ridership drop, lead/lag features capture anticipation dips, post-holiday return surges	Chen et al., 2023; Song et al., 2024
Rainfall	data.humdata.org	Weather covariate, state-level daily precipitation adds a spatially-varying external signal that influences travel mode choice	Huang et al., 2022; Cui et al., 2025
Population Density	hub.worldpop.org	Spatial context, residential density is a static proxy for transit catchment potential and latent demand at each stop	Ahmad et al., 2024; Amir et al., 2025
GADM Boundaries	gadm.org	Spatial structure, state boundary polygons are used to spatially aggregate features by administrative region and derive inter-state adjacency weights	Jin et al., 2022; Hassan et al., 2025
GTFS Static	data.gov.my	Network topology, stop counts, route density, travel-time edges quantify transit supply and accessibility per region	Keller et al., 2022; Hassan et al., 2025
OSM POIs	overpass-api.de	Land-use context, POI category counts within a 500m catchment of each stop capture activity generators that drive trip origins and destinations	Shi et al., 2024; Amir et al., 2025

Data Understanding

Attribute	Detail
Data Range	2019-01-01 to 2025-12-31 (Include MCO) 2022-01-01 to 2025-12-31 (Exclude MCO) (Lv et al., 2021; Zhao & Lan, 2025; Maria-Arribas et al., 2026)
Total Observations	2,557 daily records with MCO 1,461 daily records without MCO
Total Features	79 columns
Target Variables	13 (12 transit service lines + total ridership)
Feature Sources	8 (Ridership, fuel price, holiday/cyclical, rainfall, population density, GTFS transit network, OSM POI, GADM boundaries)

Data Understanding (con't)

Group	Count	Description
Targets	13	12 service lines + total ridership
Temporal	16	Holiday flags, lead/lag days, cyclical encoding, year, day-of-year
External	30	Fuel price levels & changes (15) + Rainfall per state (15)
Lag	3	Autoregressive lags at 7, 14, 28 days
Static	17	Population density, GTFS network states, OSM POI counts, GADM boundaries

Data Preprocessing



Per-source Cleaning

All data sources are filtered to 2019-01-01 – 2025-12-31; MCO flagging on ridership, compute total ridership; Weekly fuel price forward-fill as daily; Expand data range rows into one binary flag per calendar day, handle state-specific coverage; Retain final rainfall result; Compute state centroids; Deduplicate stops and trips; Clip density values at 99th percentile to reduce raster outliers; Deduplicate OSM element ID, map specific tags to 7 POI categories



Feature Alignment

Merged all cleaned data sources into 2019/2022-01-01 – 2025-12-31 (2,557 | 1,461 days) x 79 feature columns; Time-varying sources reindexed to daily master; Remaining gaps filled by forward-fill or linear interpolation; Static sources broadcast to every row.



Sequence Building

Sliding window sequences: (X shape: N, T_{in}, 79) – (y shape: N, 7) – for T_{in} in 14, 28, 56 look-back days.
Chronological 70/15/15 split – no shuffle
Special for graph models – build Pearson correlation adjacency on-the-fly from X_{train}

Notable Explanatory Data Analysis



MCO Collapse / Recover

Four change-point phases; 85-90% drop during MCO. Rail recovers slower than bus.



Fuel Price Elasticity

Post-MCO RON95-ridership $r \sim +0.35$; stronger above RM 2.20/L, lagged 1-3 days.



Northeast Monsoon

Wet/extreme days peak Nov-Jan; east coast > west coast. National totals mask regional effects.



Holiday Anticipation

~18% dip 1-2 days pre-holiday; spike on first workday back. Short holidays dip most.



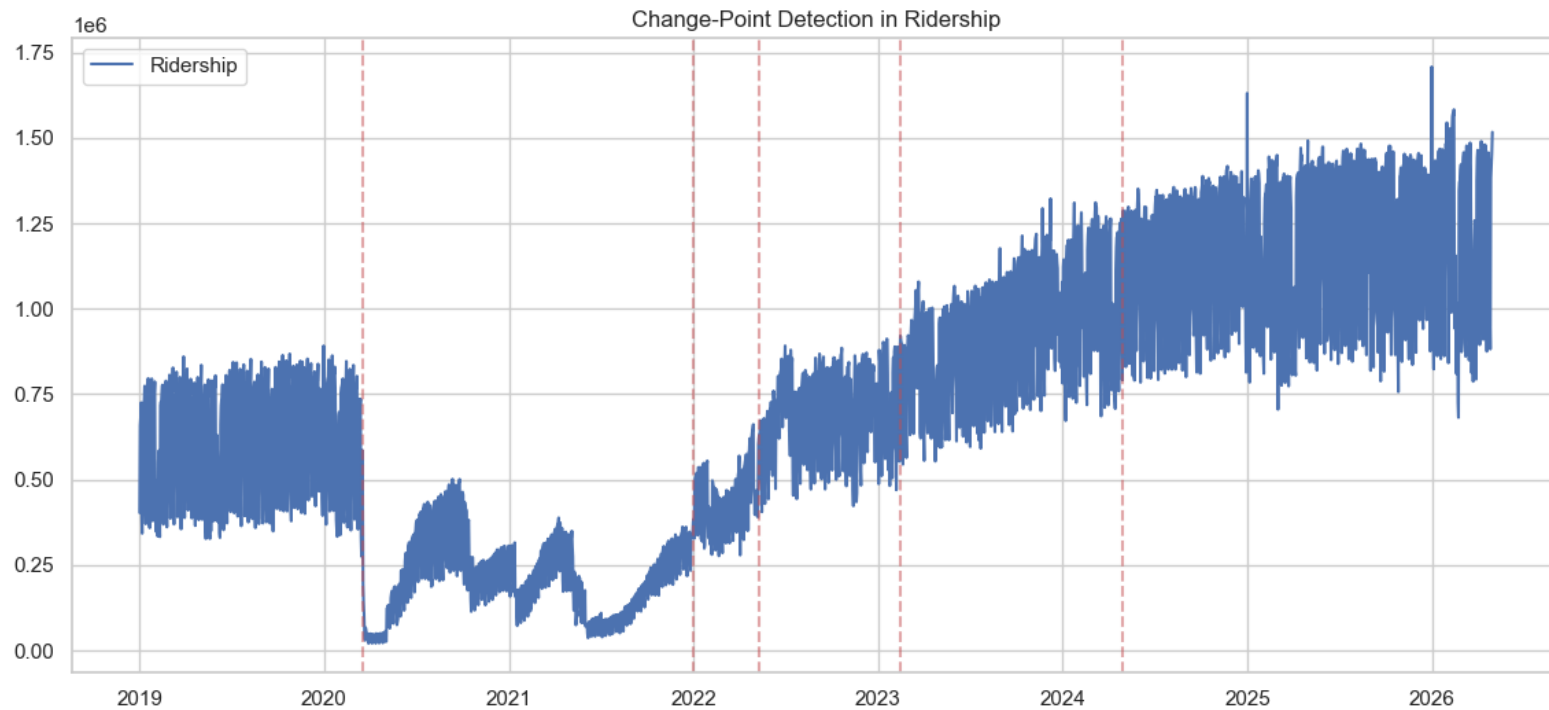
Weather-Induced Surge

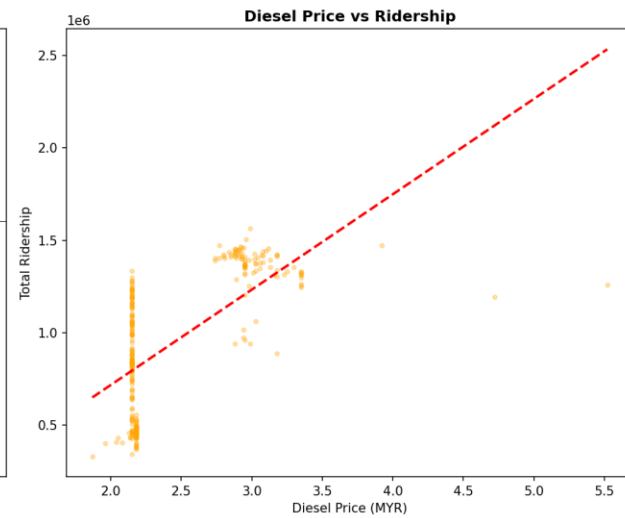
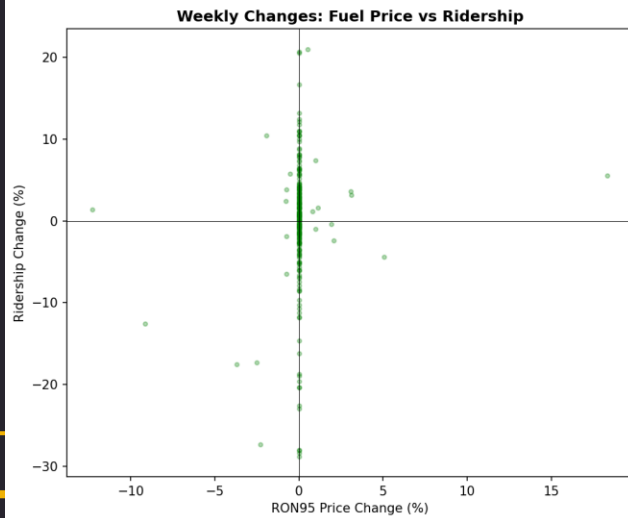
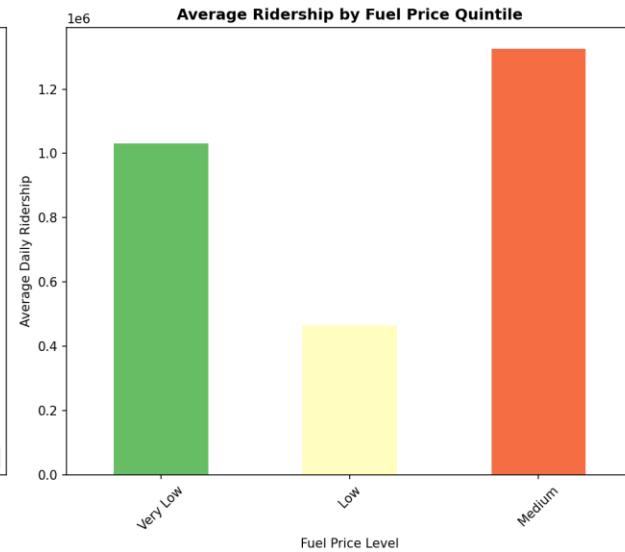
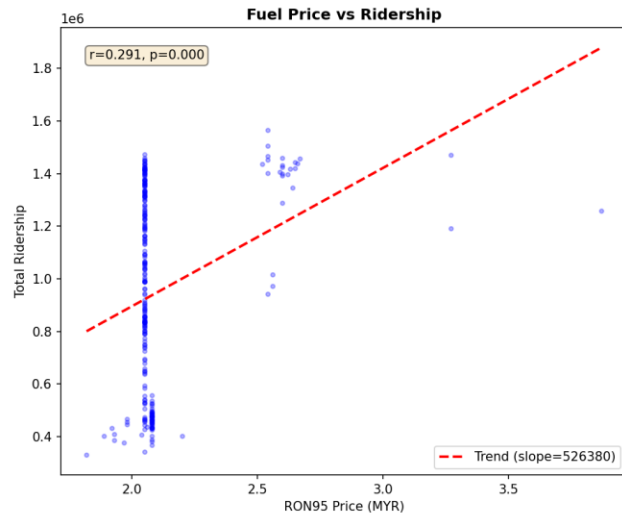
Light rain suppresses travel; heavy rain (>75th percentile) boosts ridership – modal substitution.



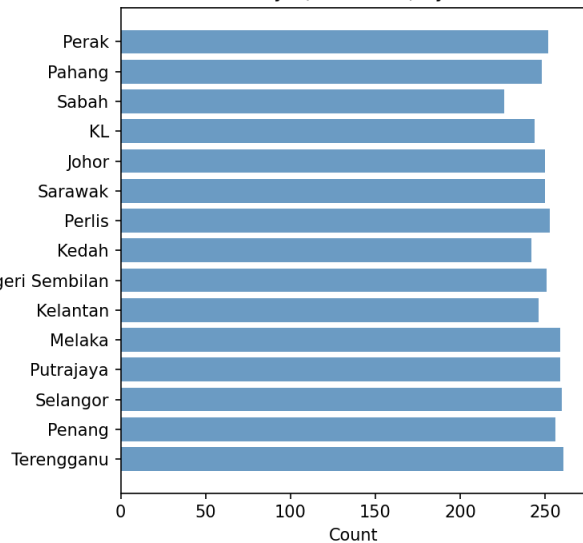
Feature Correlation

Ridership lags (7/14/28 days) explain ~72% of feature variance; static features show near-zero linear correlation.

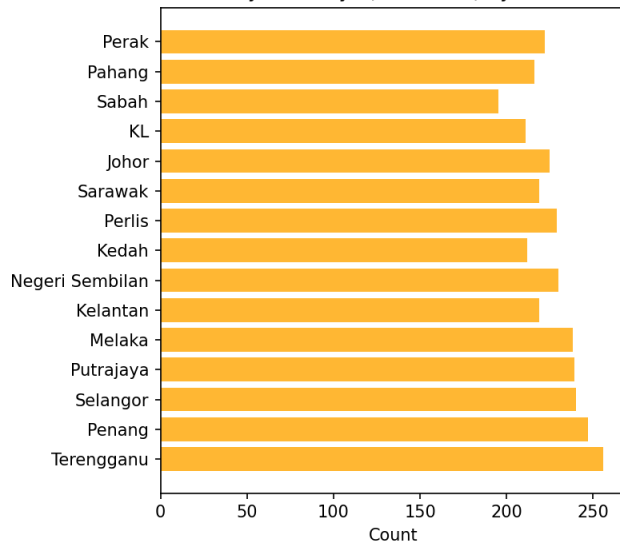




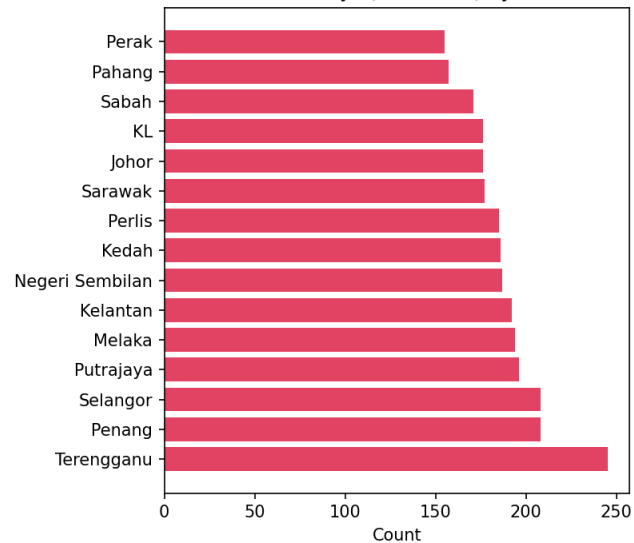
Wet Days ($\geq 10\text{mm}$) by State

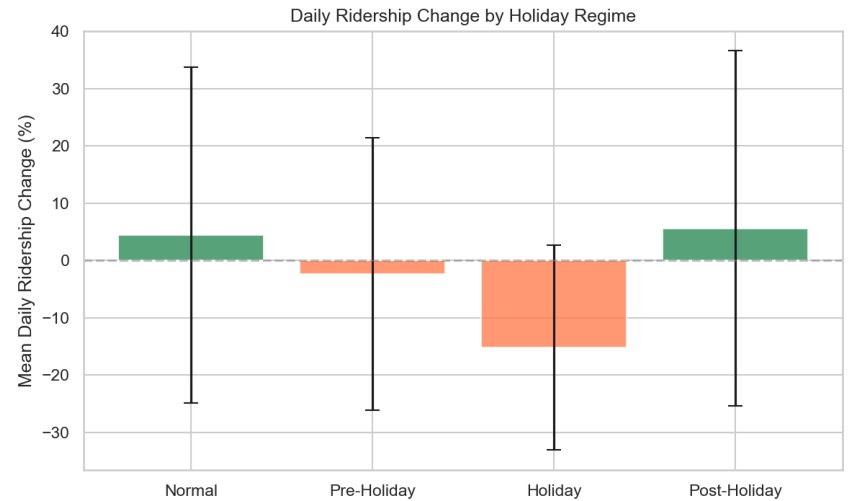
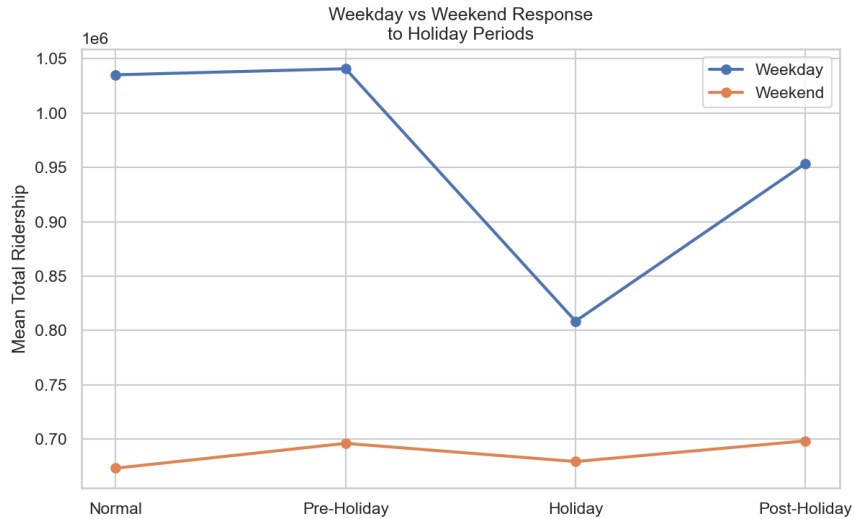
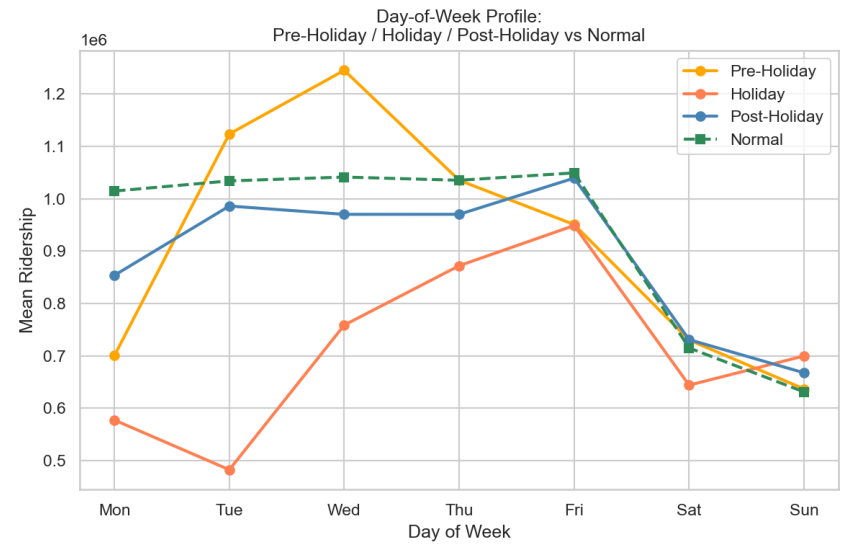
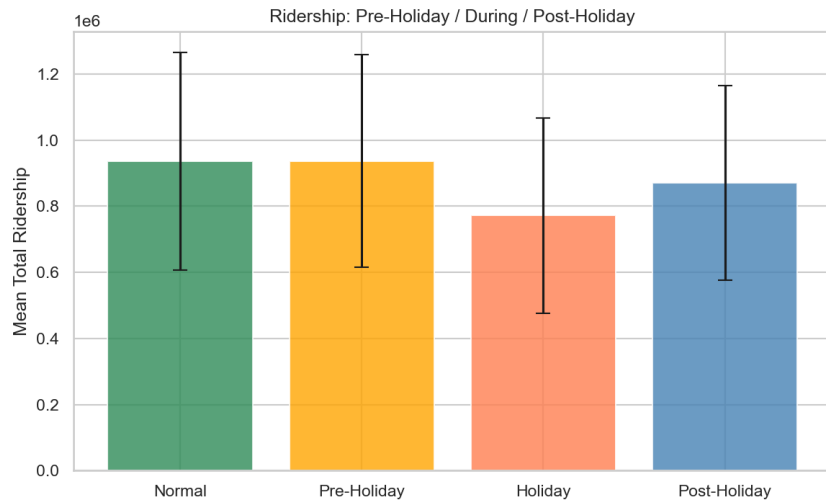


Very Wet Days ($\geq 30\text{mm}$) by State

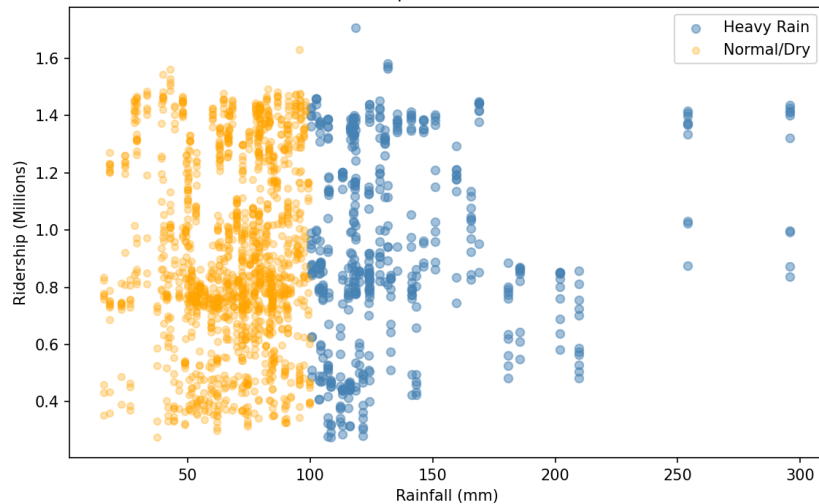


Extreme Days ($\geq 50\text{mm}$) by State

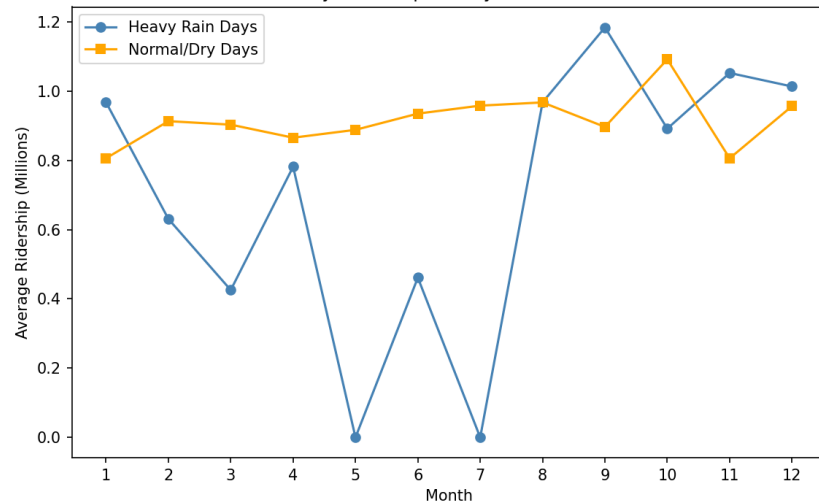




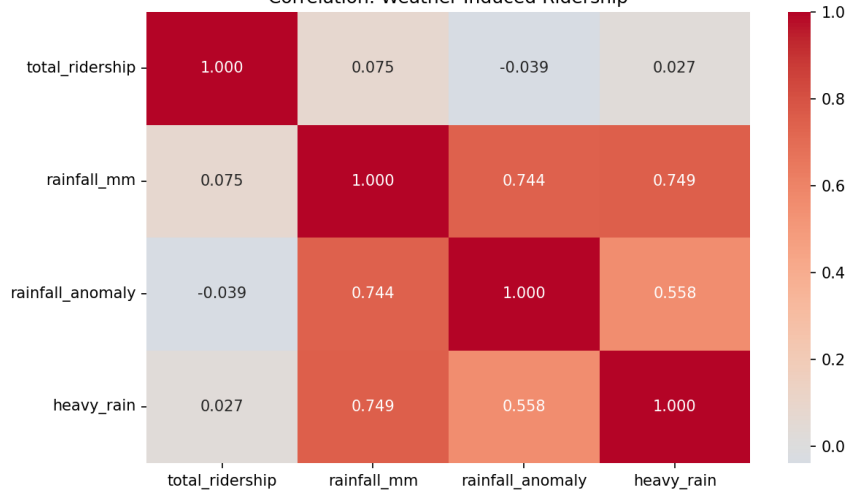
Rainfall vs Ridership: Weather-Driven Modal Shift



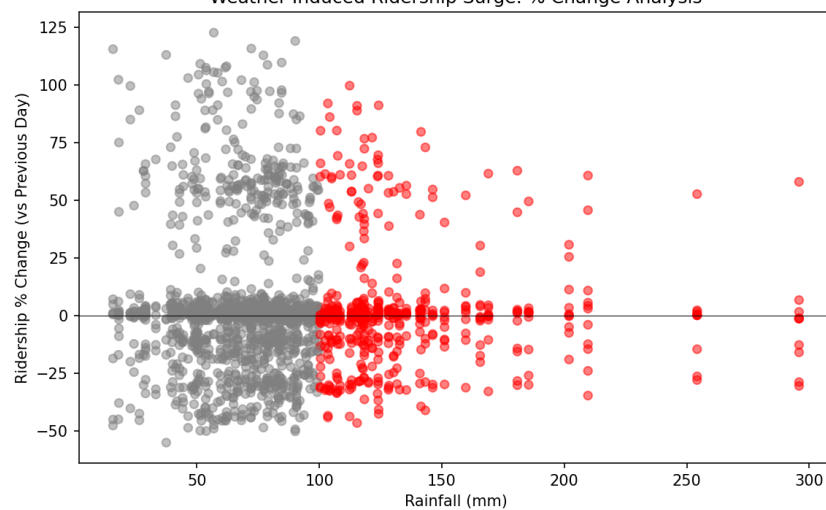
Monthly Ridership: Heavy Rain vs Normal



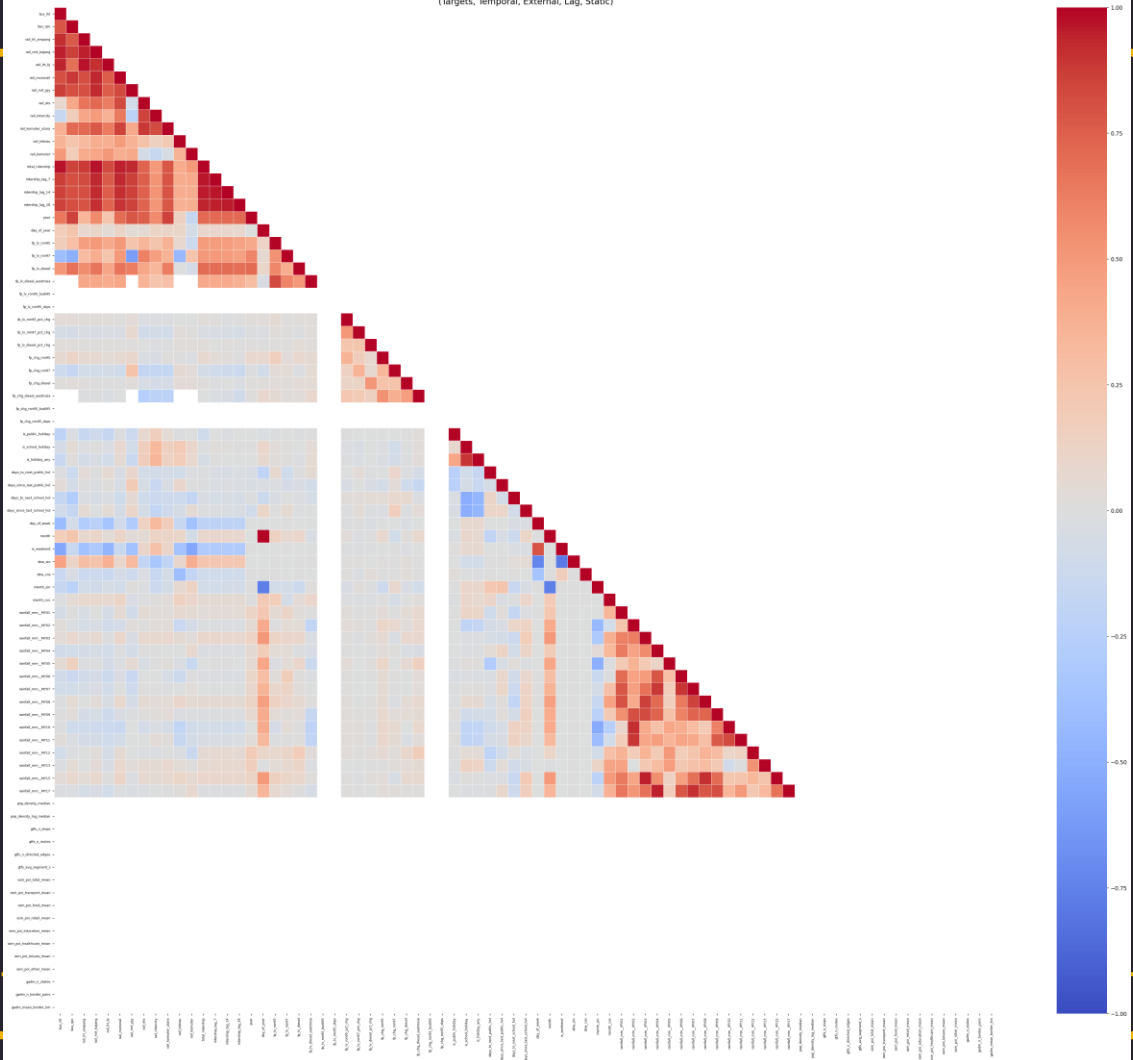
Correlation: Weather-Induced Ridership



Weather-Induced Ridership Surge: % Change Analysis



Full Demand Model Feature Correlation Matrix (79x79)
(Targets, Temporal, External, Lag, Static)





05b

Benchmark & Evaluation

Baseline LSTM

Uni-directional LSTM (64 hidden; 1 layer) to MLP

Simplest baseline, establishes performance floor, low parameter count

Single hidden vector at T is information bottleneck; no mechanism to identify which timesteps or features drive forecast



75% Combined Accuracy (CA) + 0.7 R²

CA % = MAE % + RMSE % + MAPE %



Why These Metrics?

Metrics	Role in Spatio-temporal Data	Strength	Limitations
MAE	Captures overall error across all locations and time steps	Simple, robust, same until as data	Does not emphasize large error
RMSE	Measure typical error magnitude while still penalizing spikes	Interpretable + sensitive to large deviations	Can still be dominated by others
MAPE	Compare errors across locations with different scales	Scale-independent, easy to interpret	Unstable when actual values = 0
R²	Evaluates how well the model explains spatial and temporal variance	Normalized score (up to 1), easy to compare across datasets	Heavily influenced by outliers, doesn't reflect granular local accuracy well



05c

Comparatives & Proposed Models

Spatio-temporal-Based Models

Model	Key Architecture	Strengths	Weakness
BiLSTM	Bidirectional LSTM, concatenate forward + backward hidden states	Mid-window peaks and dips are visible from both temporal directions, richer positional context	No explicit attention/pattern extraction, already near capacity ceiling at base hyperparameters.
TPA-LSTM	LSTM, Conv1d on hidden-state matrix, temporal pattern attention	Discover which recurring temporal shapes matter most, attention weights are semi-interpretable	CNN filters are fixed per training, architecture still depends on LSTM hidden vector quality
CNN-LSTM	Conv1d layers (local extraction), LSTM, three modes, sequential, parallel, augmented	Hierarchical feature extract, augmented model (CNN, LSTM + raw skip connection) generalises best by preserving direct access to input features	CNN flattens the spatial dimension of features, mode selection adds experimental complexity
CNN-BiLSTM	Conv1d, BiLSTM	Two complementary inductive biases, strong on mid-window anomalies	Higher parameter count than CNN-LSTM, requires careful regularisation to avoid overfitting
ST-LSTM	Parallel temporal stream (LSTM, spatial stream (weight-shared MLP across time), Concat	Clean conceptual separation of "when" from "which features co-activate", spatial encoder is weight-shared across timesteps	Spatial encoder mean-pools across the time dimension, losing temporal order, spatial and temporal streams are fused late without interaction during encoding

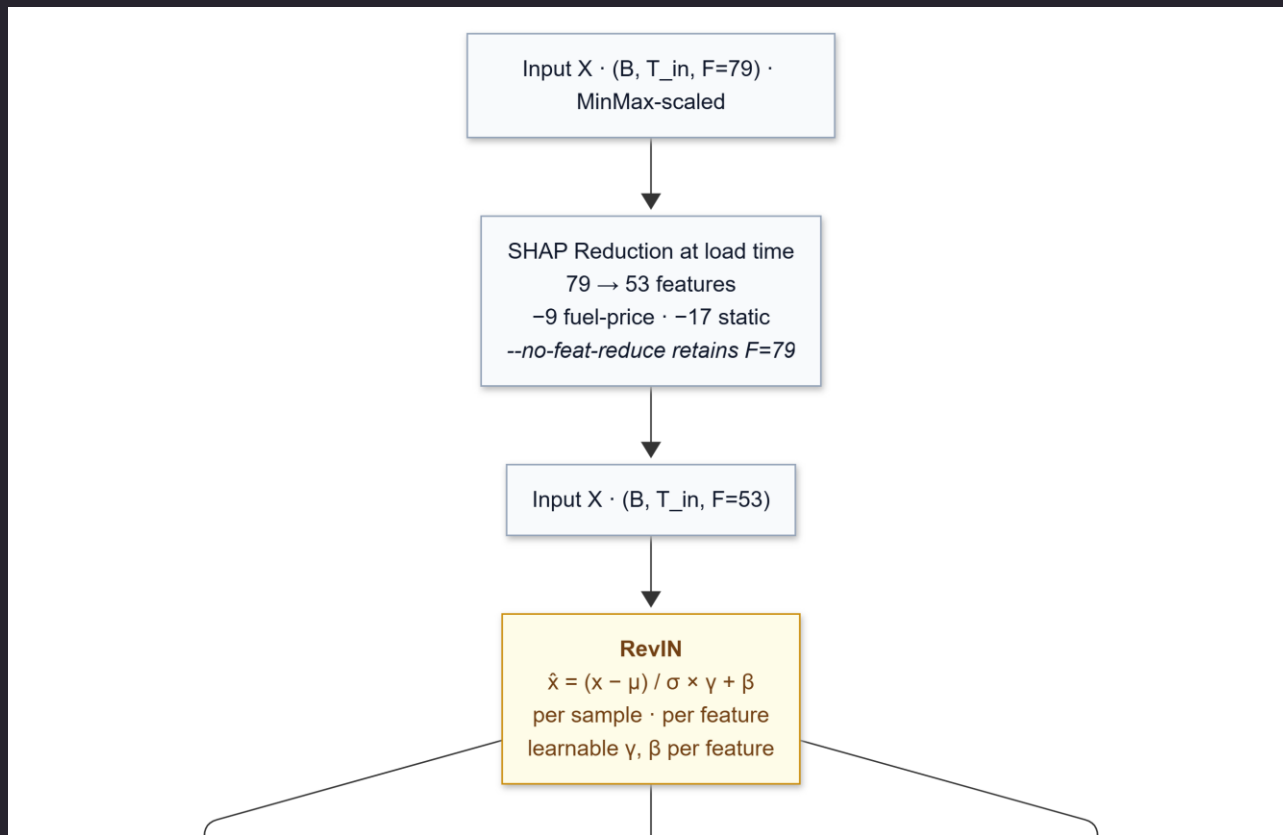
Graph-Based Models

Model	Key Architecture	Strengths	Weakness
STGCN	Interleaved temporal gated convolutions (GLU) and Chebyshev graph convolutions, fixed pre-computed Laplacian, 2 ST-Conv blocks	Computationally efficient, well validated on traffic benchmarks, interpretable static adjacency	Static graph can't adapt to changing future correlations
MTGNN	End-to-end learnable asymmetric adjacency from node embeddings, multi-scale inception blocks with dilated kernels, skip connections	No pre-computed correlation needed, asymmetric graph allows directional feature influence	Graph learning adds parameters, asymmetric adjacency can overfit the training-period topology
STSGCN	Block-tridiagonal $3N \times 3N$ spatial-temporal synchronous graph (STSG) ChebConv + GLU per layer, centre-node extraction	Unified single-pass encoding of spatial and temporal adjacency, avoids alternating ST modules, stable on long sequences	Each layer shrinks the temporal dimension by 2, limiting network depth
STFGNN	Dual static graph, spatial + temporal, per-layer learned gate blends spatial and temporal representations	Two complementary graph perspectives, gate is interpretable, LayerNorm + residual stabilises deep stacks	Over-regularisation at high dropout, high dropout coefficient suppresses useful signal
PDR-STGCN	Two-channel input: original signal + weekly lag-difference, mixed adjacency, static sym-norm Pearson + per-sample dynamic attention, learn scalar balances static vs dynamic	Only model with explicit seasonal encoding via lag-difference channel, dynamic relational graph adapts per input sample	Most complex baseline, highest number of parameters, training can be unstable

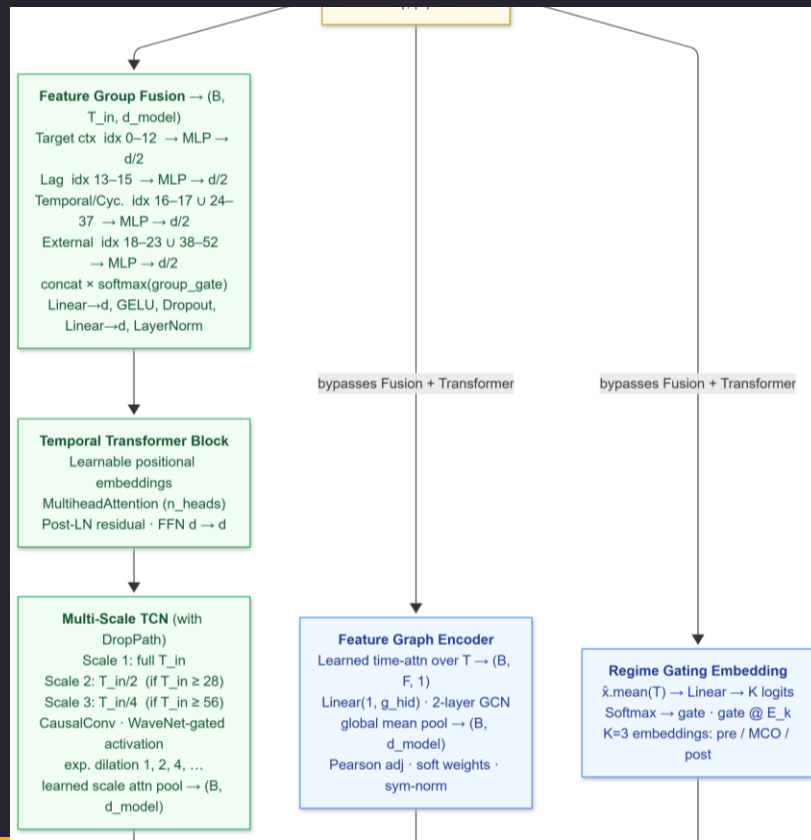
Attention-Based Models

Model	Key Architecture	Strengths	Weakness
ASTGCN	Multi-head spatial attention, Chebyshev GCN, multi-head temporal attention, repeated across n_blocks	Dual attention re-weights both spatial node relevance and temporal step relevance, transparent attention maps	Graph adjacency is still static, high parameter count per block
Autoformer	FFT-based auto-correlation discover periodic lags, explicit trend/seasonal decomposition via moving-average sub-layer throughout encoder and decoder	Sub-linear complexity, trend/seasonal decomposition aligns naturally with transit periodicity, self-correlation selects dominant lag periods automatically	FFT amplifies noise at longer sequences, sensitive to decomposition kernel width
Informer	ProbSparse self-attention, top-u queries selected by KL-divergence sparsity measure, others receive mean, sequence distilling halves T after each encoder layer, generative decoder	Graceful degradation to full attention for short sequences, generative decoder avoid autoregressive exposure bias	Sequence distilling discards some temporal resolution, many interacting hyperparameters, limited upside

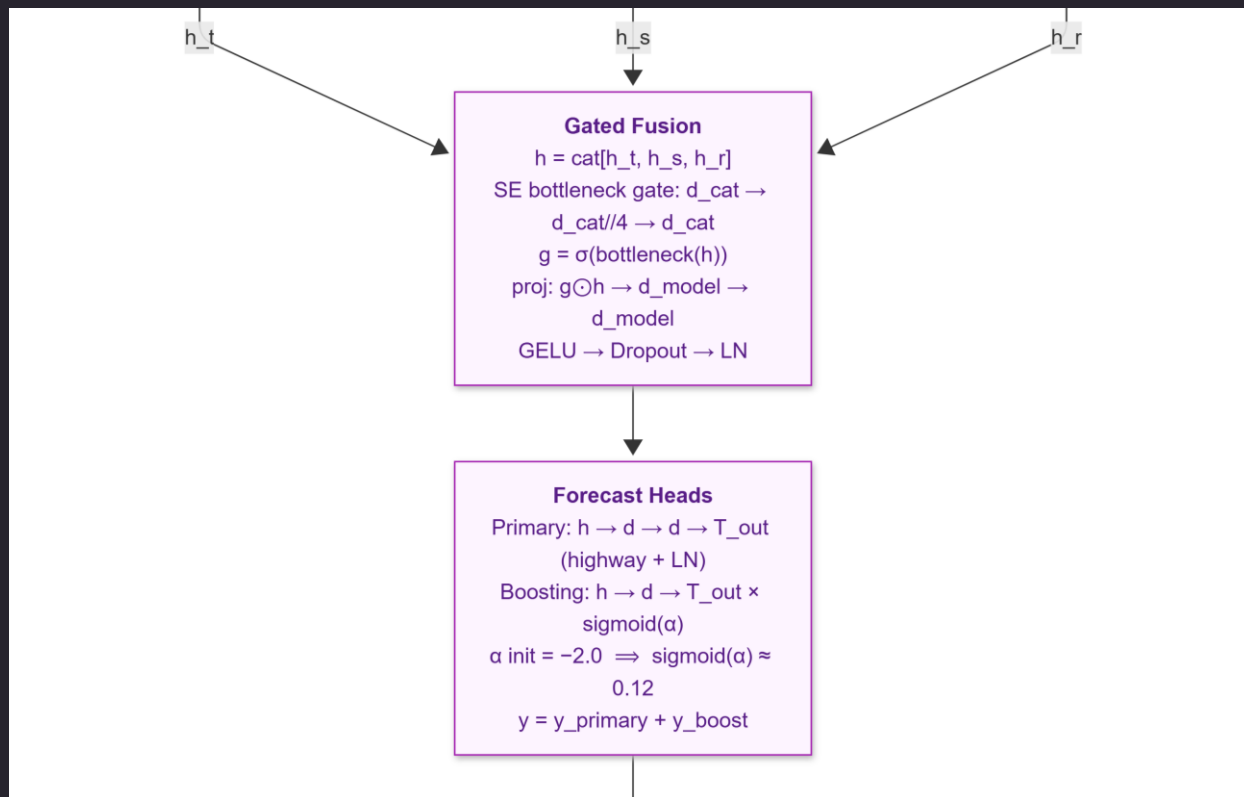
Hybrid Multi-scale Temporal Spatio-Feature (HMT-TSF) Forecaster



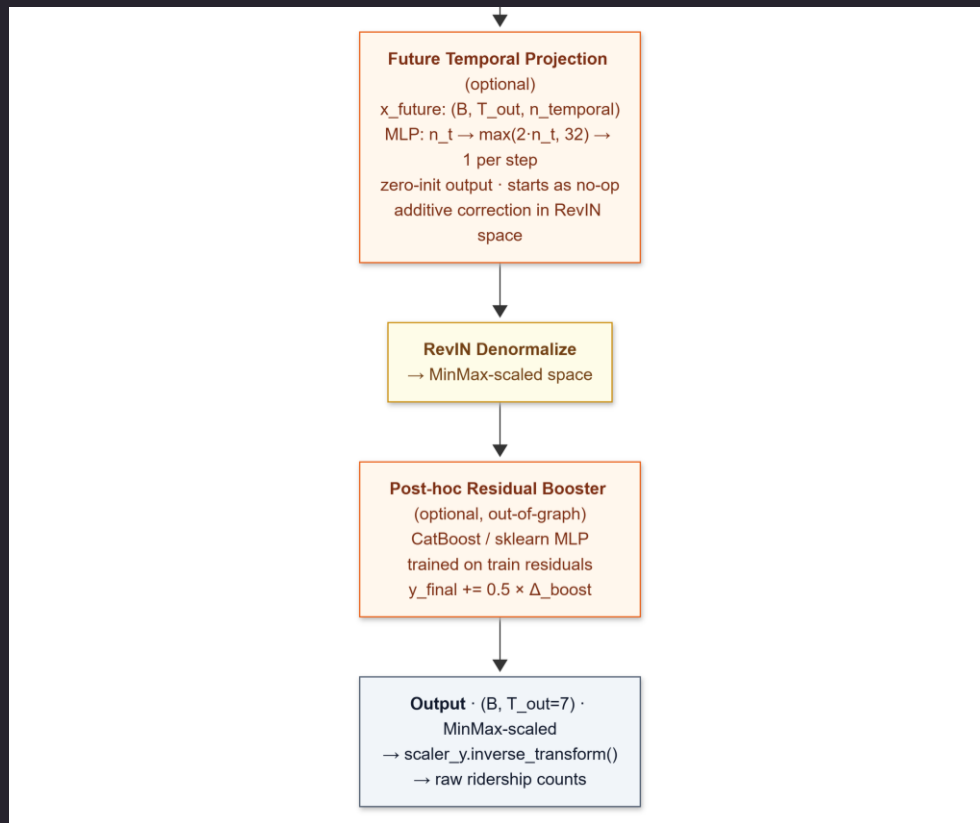
Hybrid Multi-scale Temporal Spatio-Feature (HMT-TSF) Forecaster



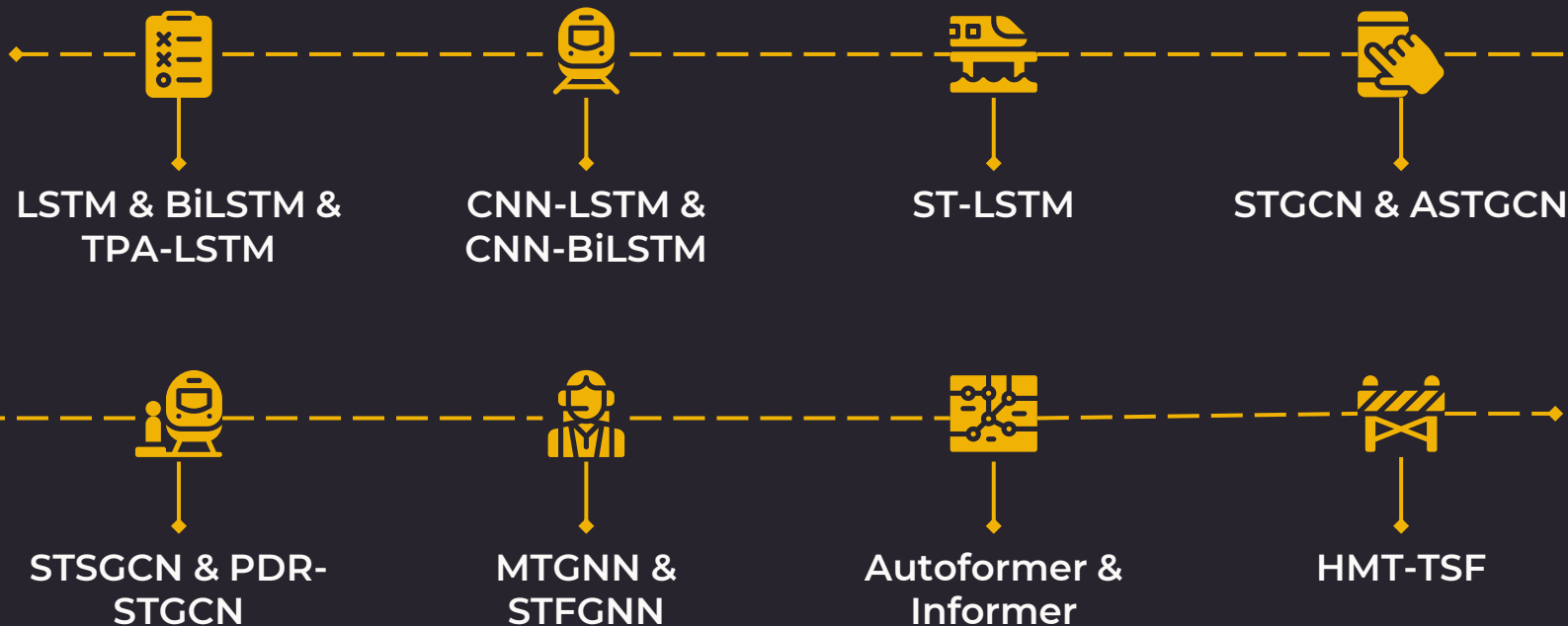
Hybrid Multi-scale Temporal Spatio-Feature (HMT-TSF) Forecaster



Hybrid Multi-scale Temporal Spatio-Feature (HMT-TSF) Forecaster



Model Selection Chronologically





06

**What I
Contribute**

Expected Outcomes

1. Provide a detailed comparative analysis evaluating multiple deep learning models across distinct lookback windows, explicitly segmented by MCO periods.
2. Introduce a superior forecasting model that outperforms existing SOTA baselines across all lookback windows and MCO phases, accompanied by a reduced feature variant according to SHAP-based feature importance ranking, to determine the effect of the removal of near-zero important spatial and temporal features toward ridership prediction.
3. Develop a functional command-line interface (CLI) inference tool capable of predicting a 7-day ridership distribution horizon across all 12 distinct transit networks utilizing historical distribution percentages using the proposed model.



Project Timeline





07

Results & Findings

Baseline Model Performance (base_nomco_lb14)

Model	Combined%	MAPE%	MAE%	RMSE%	R ²
Informer	79.13	6.59	5.53	8.76	0.772
BiLSTM	79.04	6.52	5.77	8.67	0.776
TPA-LSTM	78.67	6.62	6.04	8.67	0.776
CNN-BiLSTM	78.18	6.81	6.12	8.89	0.764
LSTM	78.13	6.71	6.18	8.98	0.760
ST-LSTM	78.01	6.93	6.27	8.79	0.770
MTGNN	77.93	6.91	6.47	8.69	0.775
CNN-LSTM	77.64	6.92	6.32	9.12	0.752
STSGCN	76.97	7.18	6.42	9.42	0.736
STGCN	76.81	7.24	6.77	9.17	0.750
ASTGCN	75.69	7.42	6.89	9.99	0.703
CNN-LSTM-Augmented	75.36	7.77	7.19	9.68	0.721
Autoformer	74.73	7.86	7.51	9.90	0.708
CNN-LSTM-Parallel	74.18	8.18	7.68	9.97	0.704
STFGNN	73.94	8.21	7.60	10.25	0.688
PDR-STGCN	73.91	8.15	7.70	10.24	0.688

Tuned Model Performance (base_nomco_lb14)

Model	Based Combined%	Tuned Combined%	Difference
Informer	79.13	79.99	+0.86
TPA-LSTM	78.67	79.95	+1.28
BiLSTM	79.04	79.44	+0.40
ST-LSTM	78.01	79.13	+1.12
ASTGCN	75.69	78.82	+3.13
LSTM	78.13	78.43	+0.30
STGCN	76.81	78.39	+1.58
CNN-LSTM-Augmented	75.36	77.10	+1.74
PDR-STGCN	73.91	75.56	+1.65
Autoformer	74.73	75.21	+0.48
MTGNN	77.93	76.40	-1.54
CNN-BiLSTM	78.18	76.12	-2.06
STSGCN	76.97	73.65	-3.33
CNN-LSTM	77.64	73.36	-4.28
CNN-LSTM-Parallel	74.18	72.53	-1.65
STFGNN	73.94	65.47	-8.47

Lookback Sensitivity

Model	lb14	lb28	lb56	Best
Informer	79.13	78.63	77.17	lb14
BiLSTM	79.04	76.97	77.09	lb14
TPA-LSTM	78.67	79.08	76.36	lb28
CNN-BiLSTM	78.18	72.65	74.72	lb14
LSTM	78.13	77.48	75.98	lb14
ST-LSTM	78.01	78.86	75.60	lb28
MTGNN	77.93	78.11	77.76	lb28
CNN-LSTM	77.64	73.90	74.33	lb14
STSGCN	76.97	75.02	75.15	lb14
STGCN	76.81	77.68	74.93	lb28
ASTGCN	75.69	76.89	75.15	lb28
CNN-LSTM-Augmented	75.36	74.73	74.04	lb14
Autoformer	74.73	73.61	73.40	lb14
CNN-LSTM-Parallel	74.18	76.55	74.33	lb28
STFGNN	73.94	61.30	43.25	lb14
PDR-STGCN	73.91	72.35	72.20	lb14

MCO Robustness (base_lb14) – Combined%

Model	nomco	mco	Difference	nomco R ²	mco R ²
Informer	79.13	73.25	-5.88	0.772	0.705
BiLSTM	79.04	72.74	-6.29	0.776	0.699
CNN-LSTM-Parallel	74.18	66.89	-7.29	0.776	0.577
TPA-LSTM	78.67	71.05	-7.61	0.764	0.669
STGCN	76.81	68.55	-8.27	0.760	0.615
MTGNN	77.93	67.33	-10.60	0.770	0.581
PDR-STGCN	73.91	63.10	-10.81	0.775	0.494
CNN-LSTM-Augmented	75.36	64.50	-10.86	0.752	0.531
ST-LSTM	78.01	66.75	-11.26	0.736	0.584
STSGCN	76.97	64.20	-12.77	0.750	0.531
ASTGCN	75.69	61.69	-14.00	0.703	0.471
CNN-BiLSTM	78.18	60.59	-17.59	0.721	0.427
LSTM	78.13	58.92	-19.21	0.708	0.390
Autoformer	74.73	51.26	-23.47	0.704	0.121
CNN-LSTM	77.64	45.58	-32.06	0.688	-0.050
STFGNN	73.94	41.00	-32.94	0.688	-0.306

HMT-TSF Model Performance (F = 79 with X_{future})

Configuration	Combined%	MAPE%	MAE%	RMSE%	R ²
nomco_lb7	84.88	4.74	4.25	6.13	0.888
nomco_lb14	85.78	4.39	3.97	5.87	0.897
nomco_lb28	84.23	4.94	4.55	6.28	0.883
nomco_lb56	82.53	5.52	5.11	6.84	0.859
nomco_lb84	77.68	6.99	6.59	8.73	0.773
mco_lb7	78.61	6.89	6.12	8.38	0.807
mco_lb14	81.99	5.68	5.18	7.15	0.859
mco_lb28	79.89	6.26	5.83	8.02	0.822
mco_lb56	79.47	6.68	5.91	7.95	0.825
mco_lb84	78.45	6.73	6.17	8.64	0.791

HMT-TSF Model Performance (F = 79 without X_{future})

Configuration	Combined%	MAPE%	MAE%	RMSE%	R ²
nomco_lb7	78.47	6.03	5.52	7.98	0.759
nomco_lb14	80.89	5.72	5.00	8.39	0.790
nomco_lb28	80.87	5.73	5.03	8.37	0.794
nomco_lb56	79.46	6.12	5.62	8.80	0.781
nomco_lb84	76.32	6.99	6.59	9.29	0.715
mco_lb7	74.60	7.41	6.79	9.20	0.697
mco_lb14	75.49	7.18	6.59	9.73	0.716
mco_lb28	74.84	7.32	6.74	9.10	0.704
mco_lb56	76.02	7.01	6.44	9.53	0.723
mco_lb84	74.82	7.12	6.58	9.50	0.705

HMT-TSF Model Performance (Feature-Reduced)

Variant	X_future	nomco_lb14 Combined%	nomco_lb14 R²	nomco_lb14 MAE	Δ vs Informer (79.13%)	mco_lb14 Combined%	MCO Δ (nomco→mco)
HMT-TSF (full)	On	85.78	0.897	49.897	+6.65	81.99	-3.79
HMT-TSF (full)	Off	80.89	0.790	62.836	+1.76	75.49	-5.40
HMT-TSF-FR	On	86.59	0.906	46.323	+7.46	78.74	-7.85
HMT-TSF-FR	Off	81.23	0.802	62.547	+2.10	74.01	-7.22



08a

Discussions

Discussion of Key Findings

1. Hybrid Multi-modal Architectures **outperform** Single-paradigm Models.
2. Test-set Accuracy conceals **Systematic Overfitting** in **recurrent baselines**.
3. **14-day** lookback window is decisively optimal **under normal conditions**.
4. **Structural Inductive Biases** determine **Robustness** under **Distributional Shifts**.
5. Learned Feature Hierarchy is **domain-consistent** and **enables Principled Feature Reduction**.





08b

Implications

Implications

1. Graph Transfer to Multivariate Ridership exposes a transfer limitation: temporal-adjacency construction breaks down at long windows.
2. Fit Diagnostics as Necessary Complement.
3. Regime Gating + RevIN for Robustness.
4. Established networks can have phased integration: runs parallel to existing planning; forecast-informed decisions.
5. Developing corridors (Johor, East Malaysia): Proof-of-Concept for demand-responsive network design, planning new services around forecasts from the outset.





08c

Limitations

Limitations

1. **Granularity** — National daily level only; no station-level or intra-day crowd management
2. **Single dataset** — Temporal stability shown but cross-network validity untested
3. **Batch-historical inputs** — Errors/delays can be imputed, not prevented upstream
4. **Limited regime diversity** — Only one shock type (MCO) informs $K = 3$ regime gating
5. **Single-seed results** — No variance/significance testing; sub-1.5% gaps are indicative only
6. **Demand-side only** — Excludes supply-side disruptions (no public dataset exists)
7. **Residual MCO leakage** — Lingers via `ridership_lag_{17,14,28}` post-MCO
8. **Confounded comparison** — Only HMT-TSF gets X_{future} calendar input; no equivalent baseline





08d

Future Works

Future Works

1. Statistical validation and symmetric input design.
2. Supply-side reliability data integration.
3. Real-time data integration.
4. Data integrity at the source.
5. Spatial and temporal disaggregation.
6. Generalisation to new service lines and networks.
7. Regime modelling beyond MCO.



THANK YOU!

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